

ABERDEEN, SD, USA

WMO: 726590

Lat: 45.443N

Lon: 98.413W

Elev: 395

StdP: 96.67

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14929

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.4	-24.4	-30.4	0.2	-26.8	-27.7	0.3	-23.9	14.8	-9.6	12.9	-9.7	3.0	320	0.670

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	32.8	22.9	31.1	22.0	29.5	21.1	24.9	30.2	23.6	29.0	22.6	27.8	6.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.9	28.5	21.9	17.4	26.9	20.7	16.1	26.0	77.9	29.9	72.8	29.0	68.3	27.8	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.3	10.1	DB	-33.3	36.2	4.2	2.6	-36.3	38.1	-38.8	39.6	-41.2	41.0	-44.3	42.9	(4)
(5)			WB	-32.5	26.8	3.6	1.2	-35.1	27.6	-37.1	28.3	-39.1	28.9	-41.7	29.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-11.0	-8.6	-1.3	6.7	13.7	19.6	22.3	20.7	16.0	7.9	-0.7	-7.9	(6)	
	DBStd	12.97	7.67	7.72	7.39	5.67	4.91	3.84	3.25	3.27	4.84	5.26	6.35	6.86	(7)	
	HDD10.0	2658	651	521	356	126	21	0	0	0	7	102	322	554	(8)	
	HDD18.3	4638	909	754	611	349	157	30	6	14	101	324	570	812	(9)	
	CDD10.0	1391	0	0	4	27	136	287	381	331	187	38	2	0	(10)	
	CDD18.3	330	0	0	0	1	14	67	128	87	31	2	0	0	(11)	
	CDH23.3	3566	0	0	2	29	223	737	1321	860	357	36	0	0	(12)	
	CDH26.7	1219	0	0	0	6	68	250	501	278	110	7	0	0	(13)	
(14) Wind	WSAvg	4.6	4.6	4.8	5.0	5.4	5.2	4.6	3.9	3.8	4.4	4.7	4.6	4.6	(14)	
(15) Precipitation	PrecAvg	548	13	14	32	53	77	96	79	61	51	41	19	11	(15)	
	PrecMax	767	36	50	111	200	187	235	248	151	140	185	76	35	(16)	
	PrecMin	254	0	1	1	3	8	12	10	6	1	0	0	0	(17)	
	PrecStd	113	9	10	25	40	43	55	48	36	38	38	18	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	11.3	21.1	26.9	31.4	34.3	35.6	33.6	32.8	27.3	19.3	10.0	(19)	
		MCWB	3.6	6.7	12.2	14.8	19.0	22.1	24.2	23.7	21.6	17.0	11.3	5.3	(20)	
	2%	DB	3.7	6.9	16.0	22.8	28.5	31.5	33.0	31.6	29.7	23.3	14.6	6.0	(21)	
		MCWB	1.5	3.9	9.9	12.6	17.6	21.5	23.9	22.8	20.4	14.9	8.9	2.4	(22)	
	5%	DB	1.6	3.9	12.4	19.7	25.8	29.5	31.1	29.6	27.3	19.9	11.1	3.2	(23)	
		MCWB	0.2	1.8	7.8	11.0	16.5	20.3	22.8	21.7	18.7	12.9	6.6	1.1	(24)	
	10%	DB	-0.3	1.8	8.8	16.8	23.0	27.6	29.5	28.0	25.0	16.9	8.2	1.1	(25)	
		MCWB	-1.4	0.4	5.3	9.7	15.0	19.3	22.0	20.7	17.8	11.5	4.7	-0.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	7.6	14.1	16.1	21.0	24.8	26.9	25.8	23.2	18.6	12.3	5.3	(27)	
		MCDB	7.0	10.8	18.2	24.6	28.6	31.1	32.1	31.0	29.6	24.3	17.7	9.6	(28)	
	2%	WB	1.6	4.0	10.6	13.4	19.2	23.2	25.4	24.1	21.7	15.9	9.4	2.7	(29)	
		MCDB	3.5	6.5	15.0	21.0	26.3	29.1	30.6	29.3	27.9	21.3	14.4	5.3	(30)	
	5%	WB	0.3	1.9	7.9	11.8	17.6	21.8	24.2	23.0	20.4	13.8	6.9	1.1	(31)	
		MCDB	1.6	3.7	12.6	18.0	23.9	27.5	29.5	28.1	25.6	18.8	11.0	3.1	(32)	
	10%	WB	-1.3	0.5	5.3	10.3	15.9	20.4	23.0	21.8	18.9	11.8	4.7	-0.2	(33)	
		MCDB	-0.3	1.9	8.8	16.1	21.7	25.9	28.1	26.5	23.6	16.3	7.8	1.2	(34)	
(35) Mean Daily Temperature Range	MDBR		10.5	10.9	10.9	12.8	12.9	12.4	12.5	13.0	13.9	12.7	11.5	10.0	(35)	
	5% DB	MCDBR	12.2	12.3	16.8	18.7	17.3	15.0	14.5	14.8	16.5	17.9	16.4	13.2	(36)	
		MCWBR	10.4	9.5	10.8	10.0	8.6	6.8	7.0	7.3	8.2	10.3	10.9	10.1	(37)	
	5% WB	MCDBR	11.9	11.7	16.2	16.4	14.8	12.7	12.7	13.2	13.4	15.9	16.1	12.5	(38)	
		MCWBR	10.2	9.1	10.8	9.2	8.0	6.8	7.0	7.2	7.5	9.7	10.8	9.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.266	0.282	0.304	0.353	0.378	0.383	0.387	0.395	0.358	0.318	0.285	0.265	(40)	
	taud		2.417	2.406	2.450	2.328	2.339	2.372	2.368	2.326	2.408	2.487	2.476	2.437	(41)	
	Ebn at Noon		857	908	934	906	890	884	876	853	859	854	828	819	(42)	
	Edh at Noon		70	87	96	119	122	119	118	118	100	80	66	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.78	2.85	3.90	4.98	5.76	6.48	6.68	5.58	4.35	2.80	1.85	1.37	(44)	
	RadStd		0.16	0.30	0.34	0.37	0.47	0.41	0.27	0.47	0.27	0.34	0.21	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page